

Ignition HMI interface to Fanuc Industrial Robots

(Fanuc R30iA controller)

Communication from Ignition HMI directly to the Fanuc robot R30iA controller requires that the robot have the “HMI Device (SNPX)” software option installed in the robot controller and communications established with the Kepware GE Fanuc Ethernet OPC-UA server.

1. FANUC ROBOT

After purchasing the “HMI Device (SNPX)” software option from Fanuc (*part number **RTL-R553-HT***) you will receive an install code called a “PAC code”. With the controllers factory PCMCIA card installed cycle power while holding the Prev & Next keys on the teach pendant. Select controlled start then from the menu select (#2) SW install, select (#3) Option, make sure the PCMCIA card is installed and press F3 done. Find the HMI Device (SNPX) option on the list and press F4 authorize and then enter the PAC code, then Press F1 install. After installation is complete press Function select (#1) cold start.

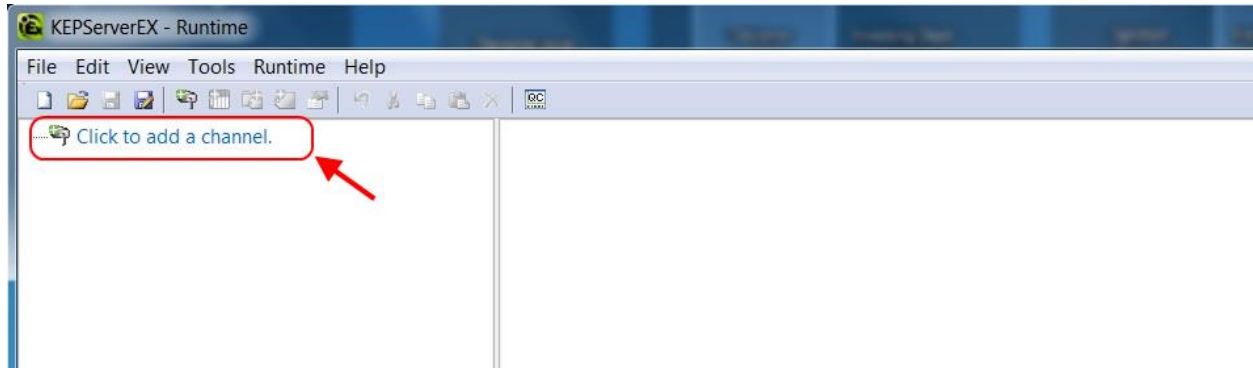
2. KEPWARE OPC SERVER

Download and install the Kepware GE Fanuc Ethernet OPC-UA server software from the Kepware website: http://www.kepware.com/Spec_Sheets/GE_Ethernet.asp

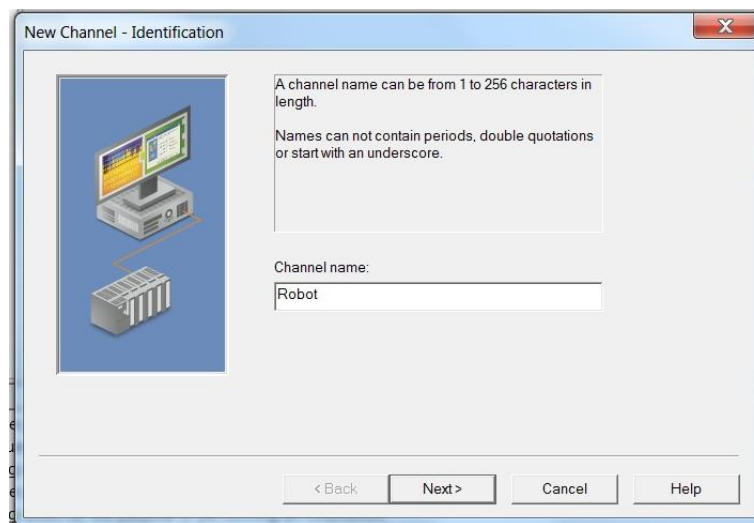


The software demo operates on a 2 hour timer. To reset the demo timer right click the Kepware icon in the system tray and select “Stop Runtime Service” then select “Start Runtime Service”.

Open the KEPServerEX configuration program and click add a channel.



Assign the channel a name and click next.

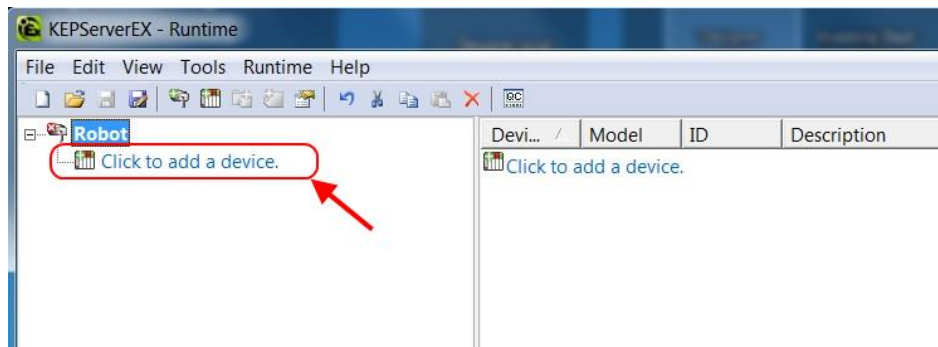


Select the GE Ethernet driver and click next.

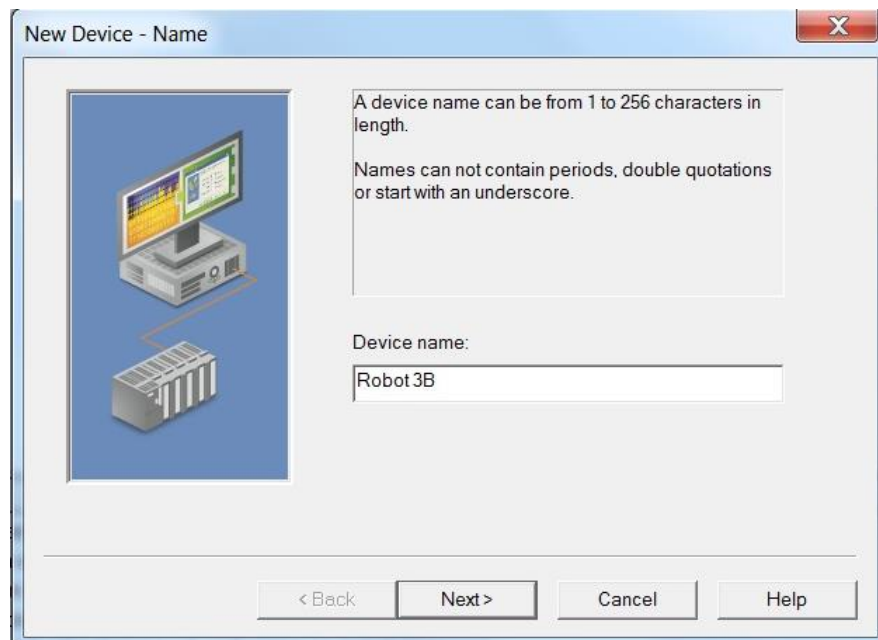


Click next for the following 3 options leaving them at their default settings (*network interface, write optimizations & float handling*) then click finish on the summary screen.

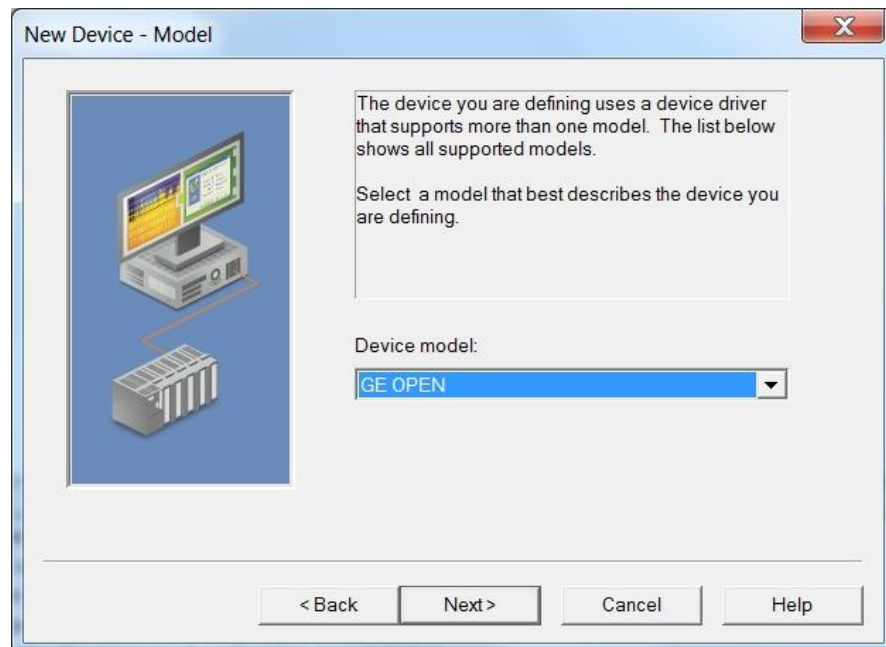
Select add a device.



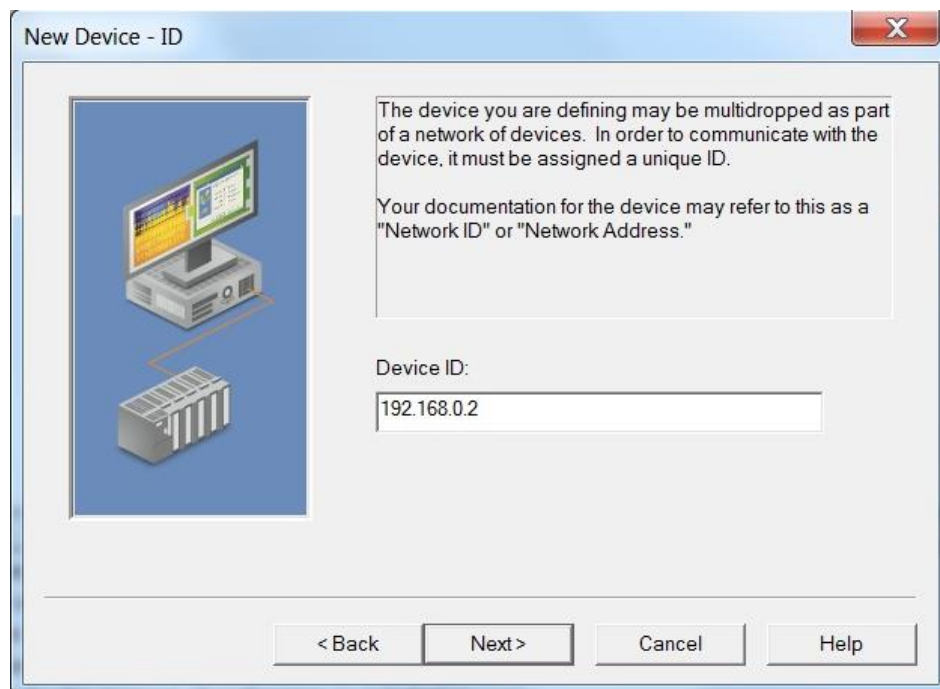
Assign the device a name.



Select the device model "GE Open".



Enter the Fanuc robot controllers IP address for controller port #2 (CO38B).

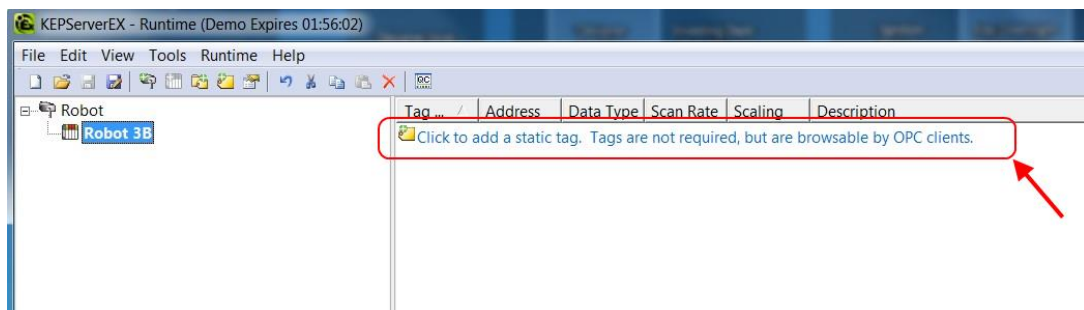


Click next for the following 7 options leaving them at their default settings (*scan mode, timing, auto-demotion, database creation, communication parameters, variable import settings & plc settings*) then click finish on the summary screen.

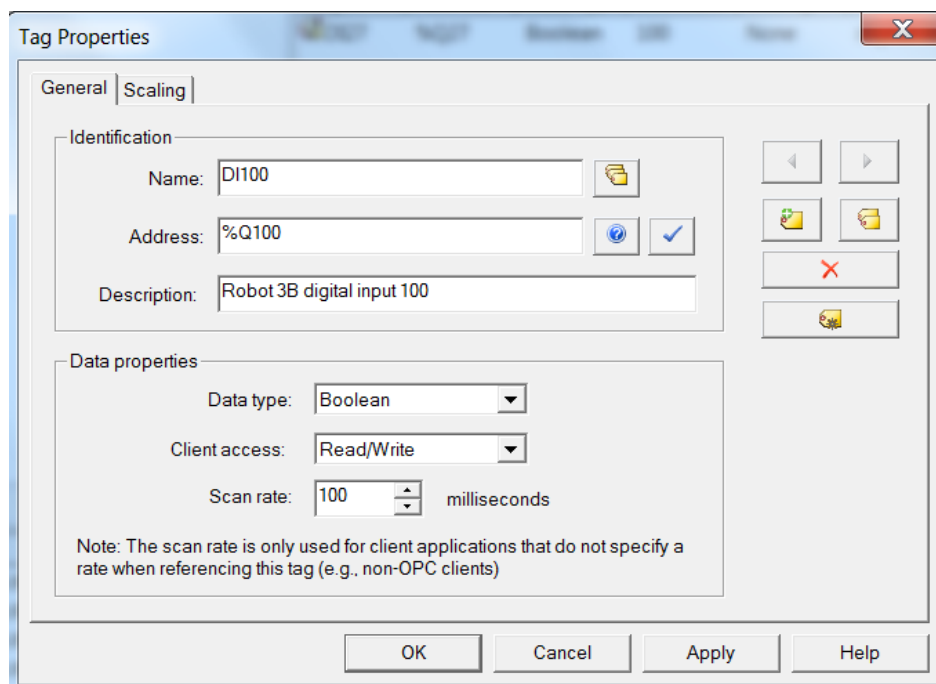
The next 4 examples will be setting up tags for digital input - outputs, Karel variables, position registers and standard registers.

Setting up a tag for a digital robot input:

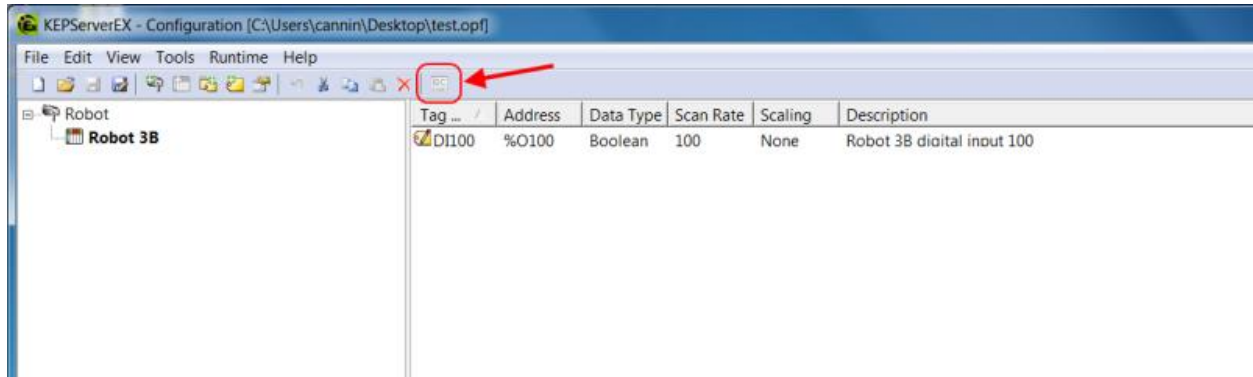
Click add a static tag.



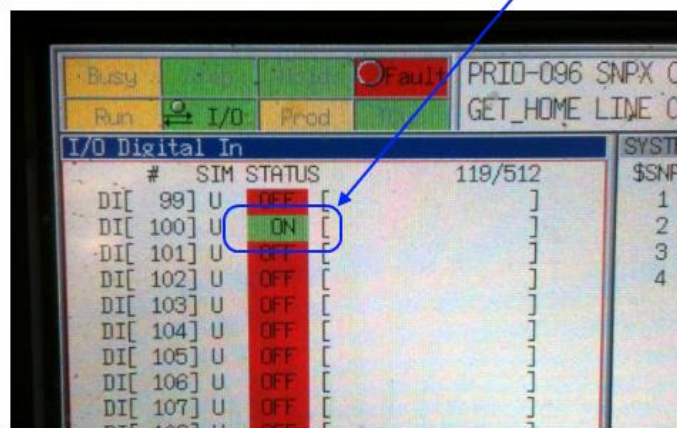
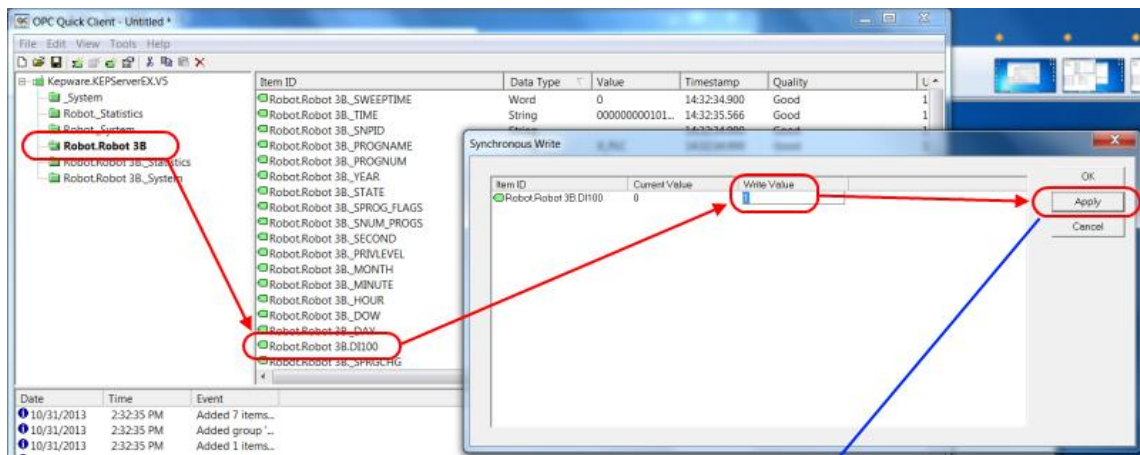
Give the tag a name, use the address prefix “%Q” for a digital input, and give the tag a description, in this example we will set up a tag for robot input 100.



To test the new tag click the Quick Client button.



Select the robot name in the left column, then right click the new tag under 'item id' and select Synchronous Write. The 'synchronous write' window will pop up. Change the value to "1" then click 'apply' and you will see the input come on in the robot teach pendant IO screen.



The table below shows the address prefixes for setting up tags for all other types of IO.

Robot controller I/O signal		PLC address	Example
Digital Input	DI[x]	%Qx	DI[1] $\Leftrightarrow$ %Q1
Digital Output	DO[x]	%Ix	DO[1] $\Leftrightarrow$ %I1
Robot Input	RI[x]	%Q(5000+x)	RI[1] $\Leftrightarrow$ %Q5001
Robot Output	RO[x]	%I(5000+x)	RO[1] $\Leftrightarrow$ %I5001
UOP Input	UI[x]	%Q(6000+x)	UI[1] $\Leftrightarrow$ %Q6001
UOP Output	UO[x]	%I(6000+x)	UO[1] $\Leftrightarrow$ %I6001
SOP Input	SI[x]	%Q(7000+x)	SI[0] $\Leftrightarrow$ %Q7001
SOP Output	SO[x]	%I(7000+x)	SO[0] $\Leftrightarrow$ %I7001
Weld Input	WI[x]	%Q(8000+x)	WI[0] $\Leftrightarrow$ %Q8001
Weld Output	WO[x]	%I(8000+x)	WO[0] $\Leftrightarrow$ %Q8001
Wire Stick Input	WSI[x]	%Q(8400+x)	WSI[0] $\Leftrightarrow$ %Q8400
Wire Stick Output	WSO[x]	%I(8400+x)	WSO[0] $\Leftrightarrow$ %Q8401
Group Input	GI[x]	%Aqx	GI[1] $\Leftrightarrow$ %AQ1
Group Output	GO[x]	%Aix	GO[1] $\Leftrightarrow$ %AI1
Analog Input	AI[x]	%AQ(1000+x)	AI[1] $\Leftrightarrow$ %AQ1001
Analog Output	AO[x]	%AI(1000+x)	AO[1] $\Leftrightarrow$ %AI1001
PMC keep relay DO[x] (x : 10001 – 10144) Ka.b		%Ix %I((a*8)+b+10001)	DO[10001] $\Leftrightarrow$ %I10001 K2.5 $\Leftrightarrow$ %I10022
PMC internal relay DO[x] (x : 11001 – 23000) Ra.b		%M(x-11000) %M((a*8)+b+1)	DO[11001] $\Leftrightarrow$ %M1 R2.5 $\Leftrightarrow$ %M22
PMCdata table GO[x] (x : 10001 – 12000) D(a*2), D((a*2)+1)		%AI(x-6000) %AI(a+4001)	GO[10001] $\Leftrightarrow$ %AI4001 D4, D5 $\Leftrightarrow$ %AI4003

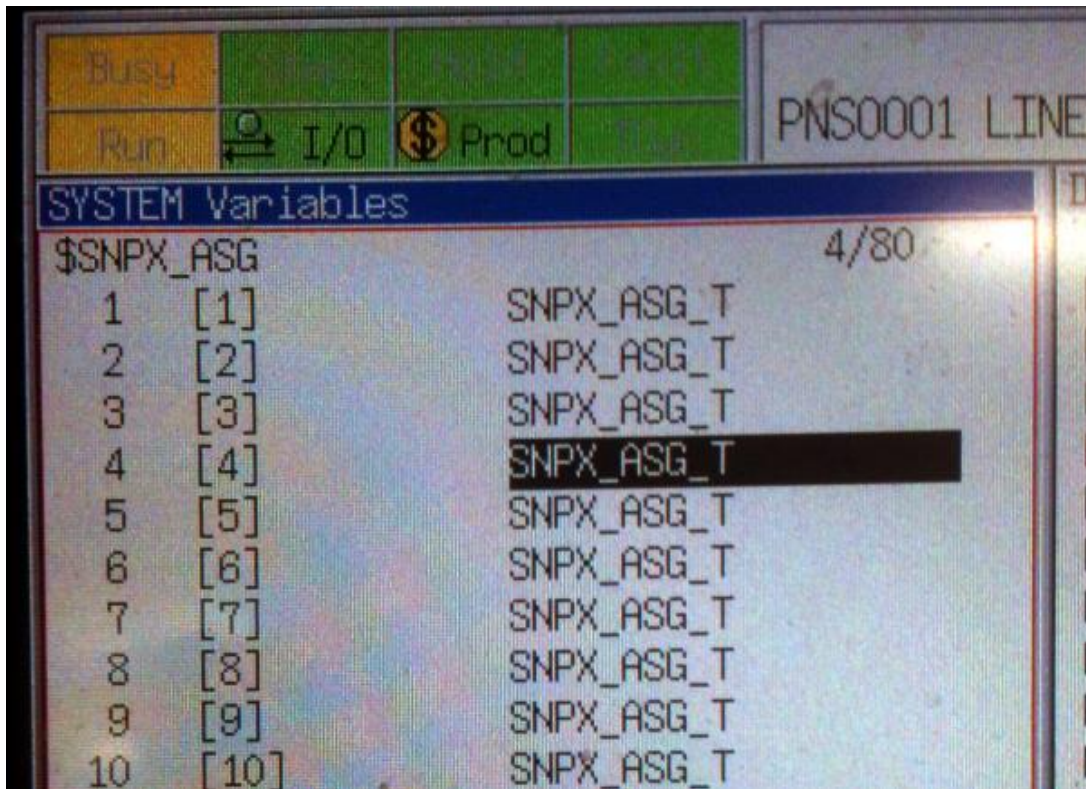
Setting up a tag for a Karel Variable:

For the next 3 examples we will be setting up tags that point to a SNPX address. We need to configure the robot controllers SNPX assignment list so it knows what to do with the '%R' information we send it with each tag.

From the robot teach pendant press 'Menu', select 'Next', select 'System', select 'Variables' then navigate to the system variable called '\$SNPX_ASG'. Here you have 80 assignment choices. Highlight '[80] of SNPX_ASG_T' and press enter.



Select one of the SNPX assignments and press enter, in this case I will use assignment #4.

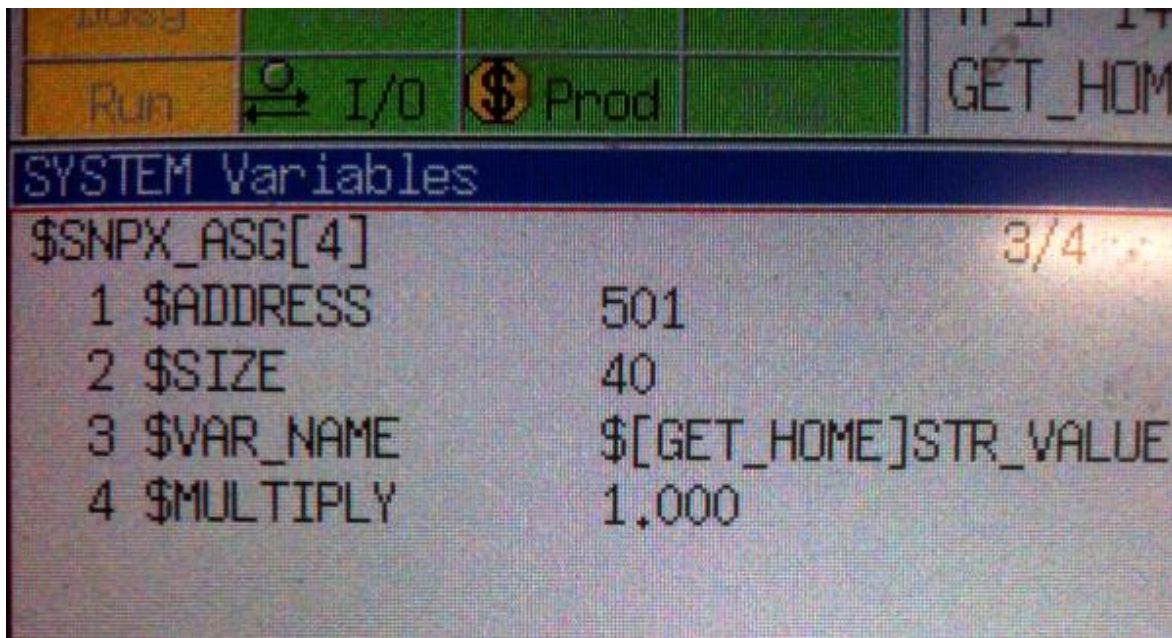


In this example we are going to write to a variable called "STR_VALUE" that is inside the Karel program called "GET_HOME". First assign a SNPX register starting address - you can choose from 1 to 16384. In this case I chose register 501. Next assign a size (meaning number of register addresses starting at 501) for a string data type we need 40 registers

NOTE: Integer, short, byte, real and boolean data types require (2) registers. Position data types require (50) registers. String data types require (40) registers.

Next assign a variable name, use the format "\$[KarelProgram]KarelVariable". In the example below it is "\$[GET_HOME]STR_VALUE". Next assign a multiply value of '1'.

NOTE: you can also write to system variables not just Karel variables using this assignment procedure.



In Kepware in the left column right click the robot and select 'new tag' and create a new tag as shown below. Give the tag a name, next assign the address range is this case we chose register 501 and a string requires 40 registers so we put in "%R501-R540 M", next enter the tag description, then set the data type to 'String' and click OK.

Tag Properties

General | Scaling

Identification

Name: Karel Variable

Address: %R501-R540 M

Description: Variable for GET_HOME, STR_VALUE

Data properties

Data type: String

Client access: Read/Write

Scan rate: 100 milliseconds

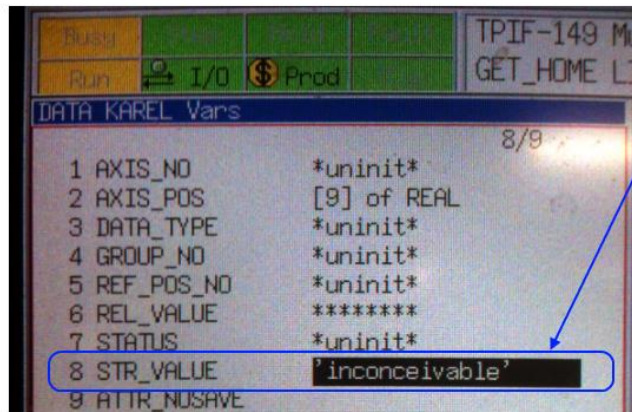
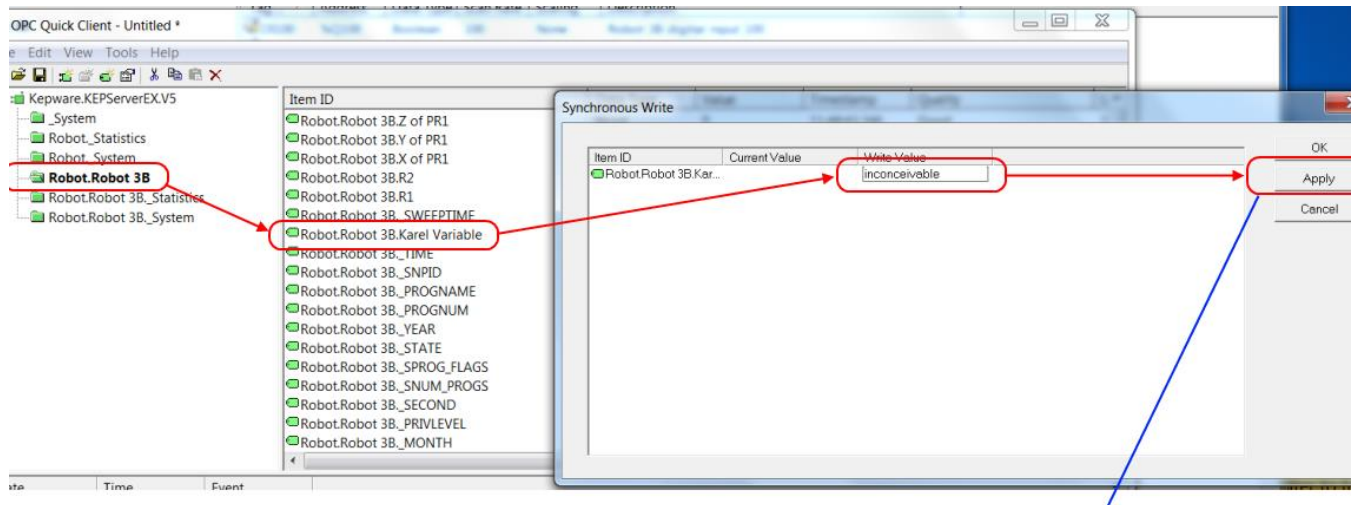
Note: The scan rate is only used for client applications that do not specify a rate when referencing this tag (e.g., non-OPC clients)

OK Cancel Apply Help

NOTE: All controller data types other than IO are passed to the robot using the '%R' prefix.

Robot controller data	PLC address	Example
Register	%Rx	R[1] ⇔ %R1

To test our new tag open the OPC quick client as before, select the robot name in the left column, then right click the new tag under 'item id' and select Synchronous Write. The 'synchronous write' window will pop up. Enter a value and then click 'apply' and you will see the variables string value change.



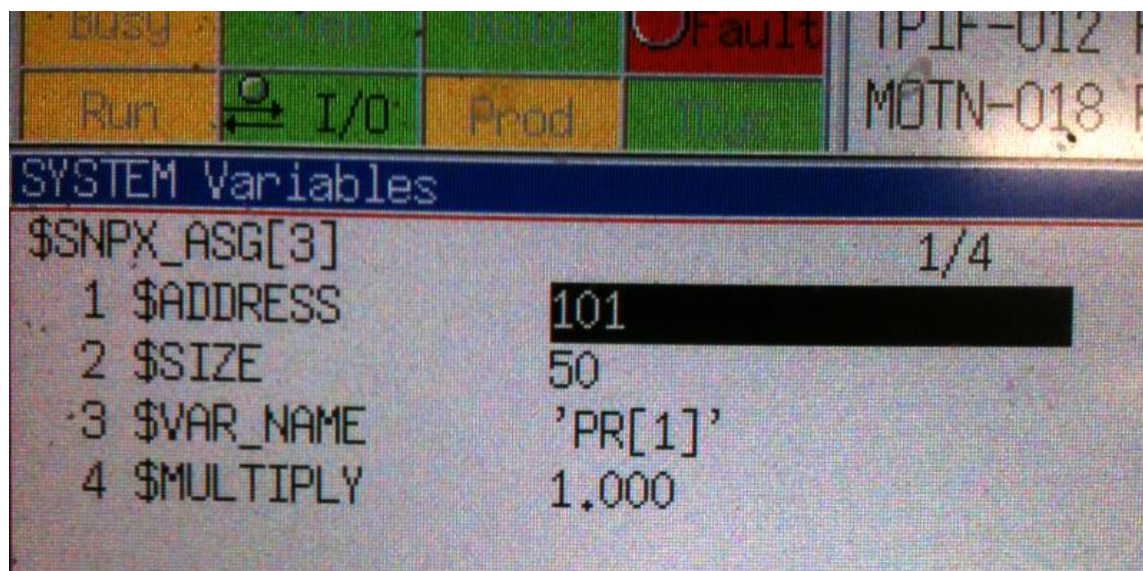
NOTE: to navigate to the robots Karel variable for this example on the teach pendant press 'Select' and select the Karel program "GET_HOME", you will see a note at the bottom of the window that says "GET_HOME is selected, next press the 'Data' key then press 'F1' TYPE and select KAREL Vars.

Setting up a tag for a Position Register:

As shown in the previous example you will need to set up a SNPX assignment. In this example I chose assignment 3. First assign a SNPX register starting address - you can choose from 1 to 16384. In this case I chose register 101. Next assign a size (meaning number of register addresses starting at 101) for a position register data type we need 50 registers.

NOTE: Integer, short, byte, real and boolean data types require (2) registers. Position data types require (50) registers. String data types require (40) registers.

Next assign a variable name as shown below. For position registers use the format "PR[#]", in this case "PR[1]" for position register 1. Next assign a multiply value of '1'.



In Kepware in the left column right click the robot and select 'new tag' and create a new tag as shown below. In this example we will create a tag for the 'X' value of position register #1. Give the tag a name, next assign the address, in this case we chose SNPX register 101 as the starting register of the block of 50 required registers. On the next page you will see a table showing how these 50 registers are broken up – in this example we only need the first 2 of the 50 registers to set the 'X' value of the position register and we only need to specify the starting SNPX register so we put in "%R101" , next enter the tag description and click OK.

Tag Properties

General | Scaling

Identification

Name: X of PR1

Address: %R101

Description: X value for position register #1

Data properties

Data type: Default

Client access: Read/Write

Scan rate: 100 milliseconds

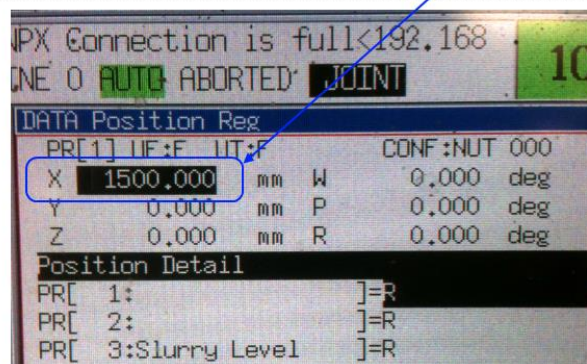
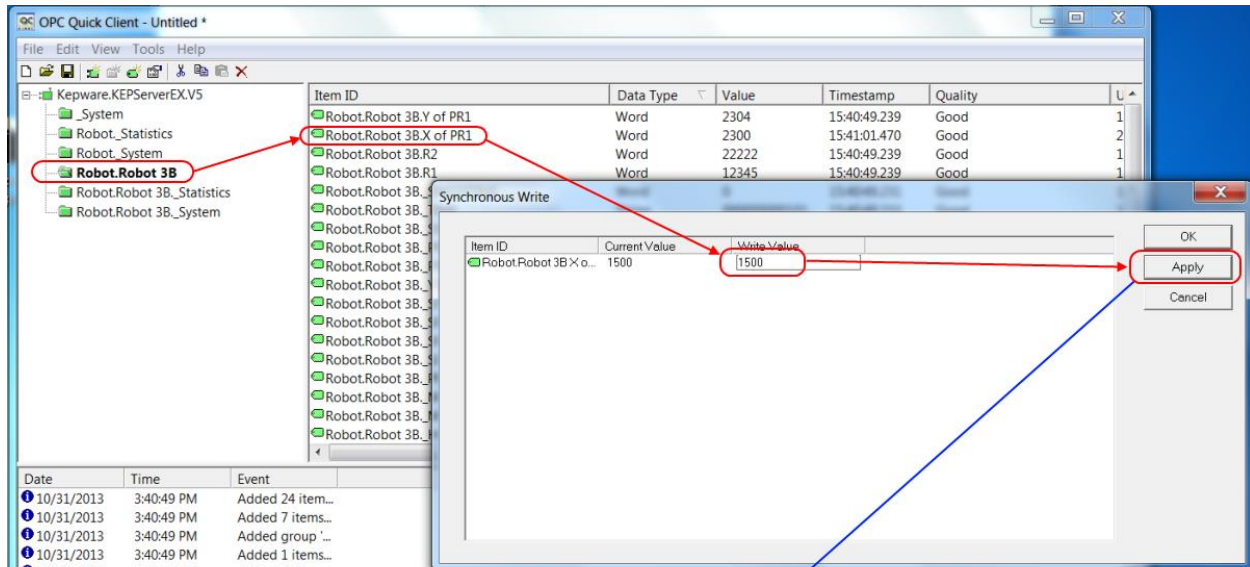
Note: The scan rate is only used for client applications that do not specify a rate when referencing this tag (e.g., non-OPC clients)

OK Cancel Apply Help

This table shows how the block of 50 registers are parsed out for assigning values to components of a position register. In the above example we are writing to the 'X' value so we used the first 2 registers of our block which was "%R101". If we wanted to create a tag to write to the 'Y' value of the position register the tags address would be "%R103" and the 'Z' value would be "%R105" and so on.

%R address	Explanation		Influence by \$MULTIPLY
Cartesian coordinate data			
1-2	X	32-bit signed integer or real number (mm)	Influenced
3-4	Y	32-bit signed integer or real number (mm)	Influenced
5-6	Z	32-bit signed integer or real number (mm)	Influenced
7-8	W	32-bit signed integer or real number (deg)	Influenced
9-10	P	32-bit signed integer or real number (deg)	Influenced
11-12	R	32-bit signed integer or real number (deg)	Influenced
13-14	E1	32-bit signed integer or real number (mm, deg)	Influenced
15-16	E2	32-bit signed integer or real number (mm, deg)	Influenced
17-18	E3	32-bit signed integer or real number (mm, deg)	Influenced
19	FLIP	16-bit signed integer (1:Flip, 0:Non flip)	Not influenced
20	LEFT	16-bit signed integer (1:Left, 0:Right)	Not influenced
21	UP	16-bit signed integer (1:Up, 0:Down)	Not influenced
22	FRONT	16-bit signed integer (1:Front, 0:Back)	Not influenced
23	TURN4	16-bit signed integer (-128~127)	Not influenced
24	TURN5	16-bit signed integer (-128~127)	Not influenced
25	TURN6	16-bit signed integer (-128~127)	Not influenced
26	VALIDC	16-bit signed integer (*1)	Not influenced
Joint data			
27-28	J1	32-bit signed integer or real number (mm, deg)	Influenced
29-30	J2	32-bit signed integer or real number (mm, deg)	Influenced
31-32	J3	32-bit signed integer or real number (mm, deg)	Influenced
33-34	J4	32-bit signed integer or real number (mm, deg)	Influenced
35-36	J5	32-bit signed integer or real number (mm, deg)	Influenced
37-38	J6	32-bit signed integer or real number (mm, deg)	Influenced
39-40	J7	32-bit signed integer or real number (mm, deg)	Influenced

To test the new tag open the OPC quick client as before, select the robot name in the left column, then right click the new tag under 'item id' and select Synchronous Write. The 'synchronous write' window will pop up. Enter a value and then click 'apply' and you will see the position registers 'X' value change.

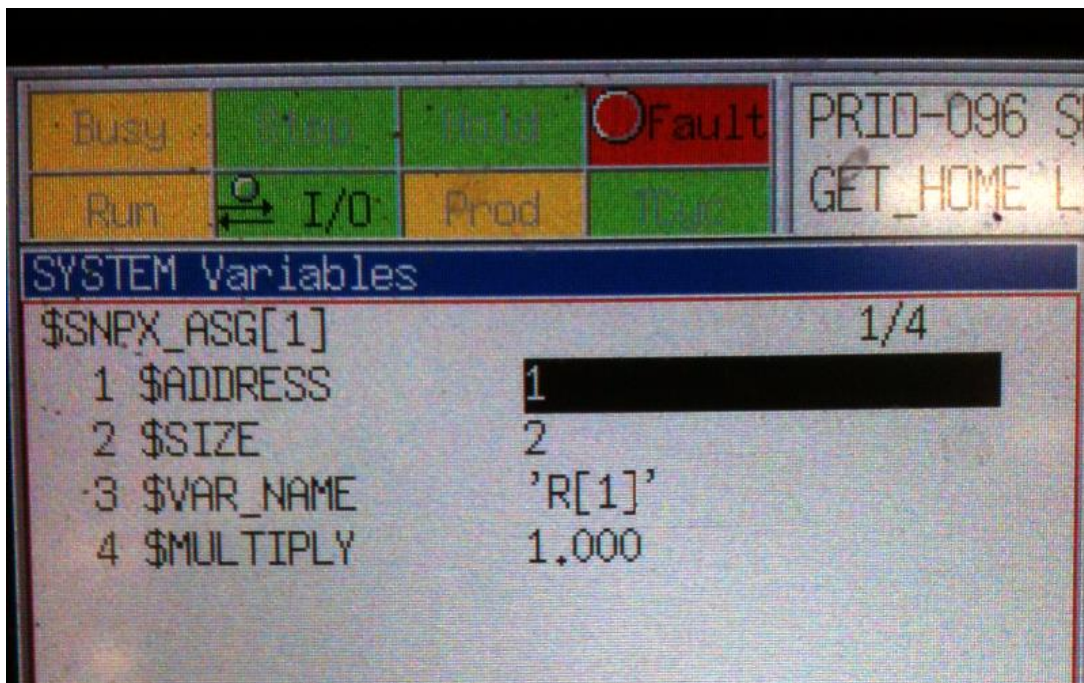


Setting up a tag for a Standard Register:

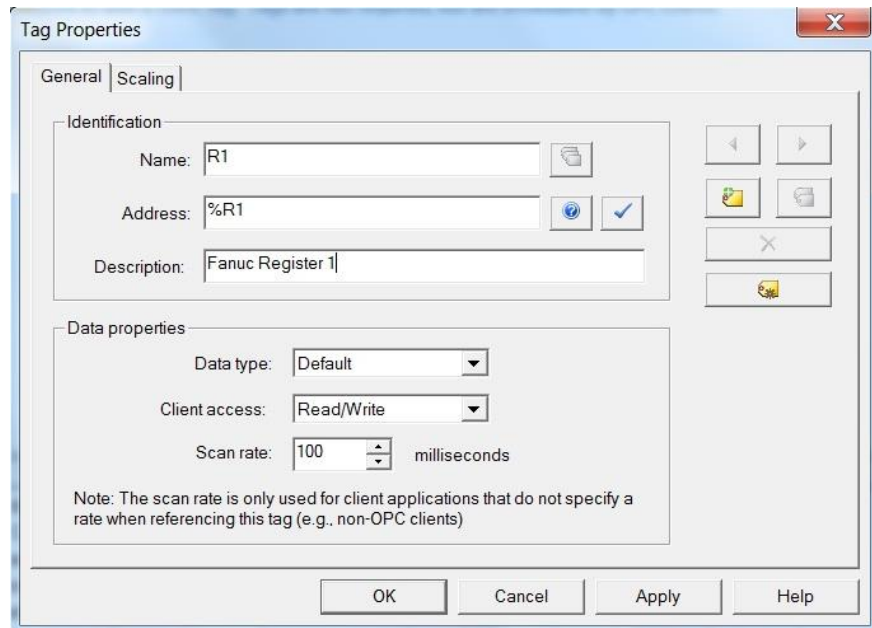
As shown in the previous example you will need to set up a SNPX assignment. In this example I chose assignment 1. First assign a SNPX register starting address - you can choose from 1 to 16384. In this case I chose register 1. Next assign a size (meaning number of register addresses starting at register 1) for an integer data type we need 2 registers.

NOTE: Integer, short, byte, real and boolean data types require (2) registers. Position data types require (50) registers. String data types require (40) registers.

Next assign a variable name as shown below. For a standard registers use the format "R[#]", in this case "R[1]" for register 1. Next assign a multiply value of '1'.



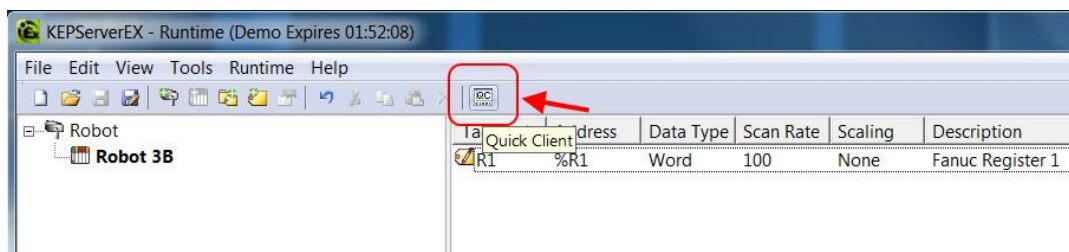
In Kepware in the left column right click the robot and select 'new tag' and create a new tag as shown below. Assign the tag a name, next assign an address, we only need to specify the starting SNPX register. Assign a description and click OK.



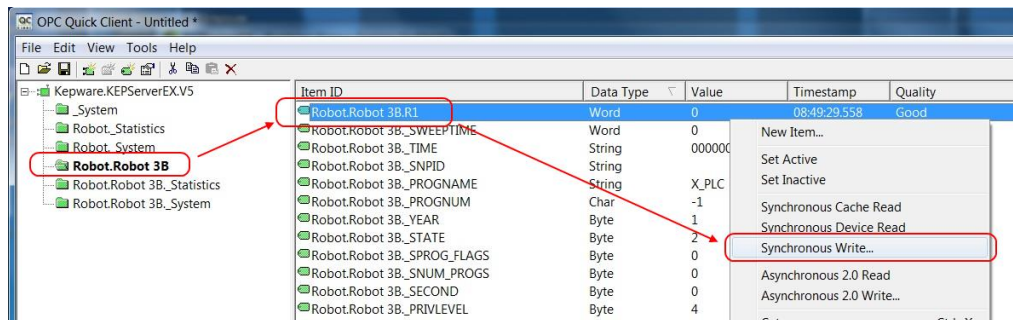
The 'Tag Properties' dialog box is shown with the 'General' tab selected. It contains two main sections: 'Identification' and 'Data properties'. In the 'Identification' section, the 'Name' field is set to 'R1', the 'Address' field is set to '%R1', and the 'Description' field is set to 'Fanuc Register 1'. In the 'Data properties' section, the 'Data type' is set to 'Default', the 'Client access' is set to 'Read/Write', and the 'Scan rate' is set to '100' milliseconds. A note at the bottom states: 'Note: The scan rate is only used for client applications that do not specify a rate when referencing this tag (e.g. non-OPC clients)'. The dialog has 'OK', 'Cancel', 'Apply', and 'Help' buttons at the bottom.

Section	Field	Value
Identification	Name	R1
	Address	%R1
	Description	Fanuc Register 1
Data properties	Data type	Default
	Client access	Read/Write
	Scan rate	100 milliseconds

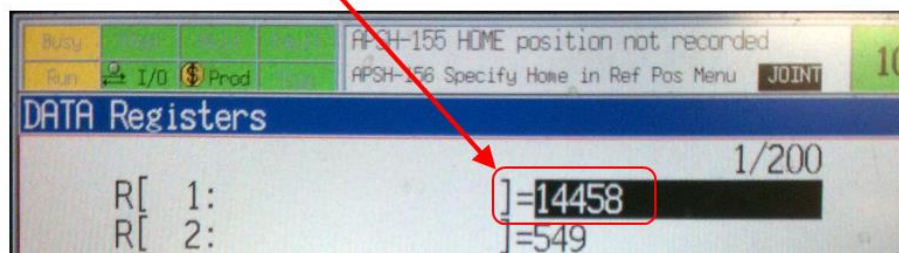
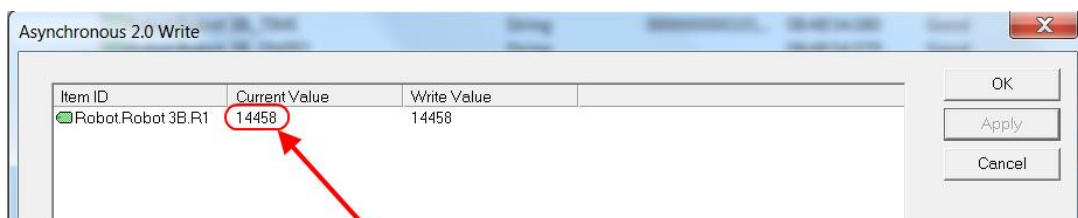
To test the new tag click the Quick Client button.



Select the robot name then right click the new tag and select Synchronous Write.



Enter a value and click Apply. The value on the robot teach pendant should change to the chosen value.



NOTE: Refer to Chapter 6 of the Fanuc Cimplicity HMI manual for more details on address assignments.

3. IGNITION SETUP

In the Ignition gateway select the configure tab then click Servers.

The screenshot shows the Ignition Gateway web interface. The 'Configure' tab is selected in the top navigation bar. The left sidebar shows the 'System' menu with 'Status' highlighted. The 'System Status' page displays various system properties:

System Properties	
Context State	Running
Redundancy Mode	Independent
Redundancy State	Active
Version	7.5.5 (b1255)
Java Version	Sun Microsystems Inc. 1.6.0_31
CPU %	0.3
Memory (used/max)	787 mb / 2.0 gb
Uptime	17 days, 13 hours, 47 minutes, 45 seconds
Timezone	America/Los_Angeles [GMT-8:00]
Locale	en_US

A yellow banner at the bottom indicates: **Trial time remaining: 1:59:11**. One or more modules are running in trial mode.

Click Create new OPC Server Connection.

The screenshot shows the 'OPC Server Connections' page. The left sidebar shows the 'Configuration' menu with 'Servers' highlighted. The page displays a table of OPC server connections:

Name	Type	Description	Status	
Ignition OPC-UA Server	OPC-UA Connection	The default connection to Ignition's OPC-UA server.	Connecting	edit delete
Production	Production_ServerType?	Production OPC-UA connection.	Faulted	delete

A red arrow points to the 'Create new OPC Server Connection...' link below the table. A note at the bottom states: **Note:** For details about a connection's status, see the [OPC Connection Status](#) page.

Select OPC-UA Connection and click next.

System

- Status
- Licensing
- Backup/Restore
- Console
- User Manual

Configuration

- Projects
- Modules
- Gateway Settings
- Redundancy

Add OPC Server Connection Step 1: Choose Type

☐ **OPC-DA COM Connection**
Provides access to legacy COM-based OPC-DA servers. Supports OPC-DA versions 2 and 3.

☒ **OPC-UA Connection**
Connection to an OPC-UA server.

Next >

Assign the new connection a name and description. Assign the host ip address (*use 127.0.0.1 if on the local machine*). Use the Kepware default port 49320, select “Basic 128Rsa15” for security policy, select “SignAndEncrypt” for security mode, leave username and password blank.

System

Status

Licensing

Backup/Restore

Console

User Manual

Configuration

Projects

Modules

Gateway Settings

Redundancy

Databases

Connections

Drivers

Store and Forward

OPC Connections

→ Servers

Quick Client

OPC-UA

Certificates

Settings

Devices

Mobile

Settings

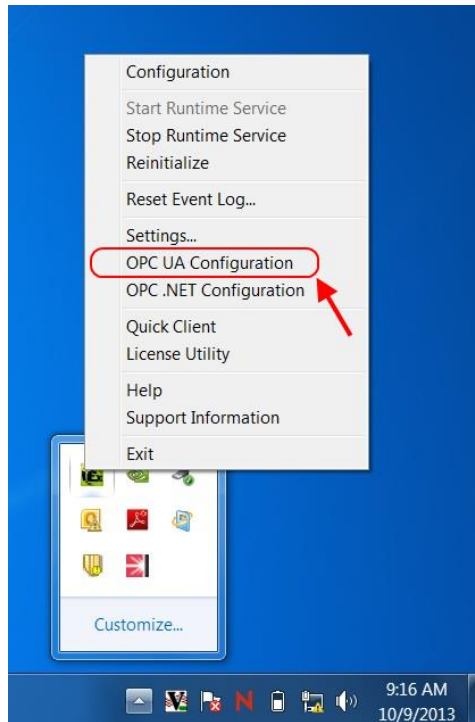
Edit OPC Server Connection

Main	
Name	Kepware
Description	GE Fanuc ethernet demo

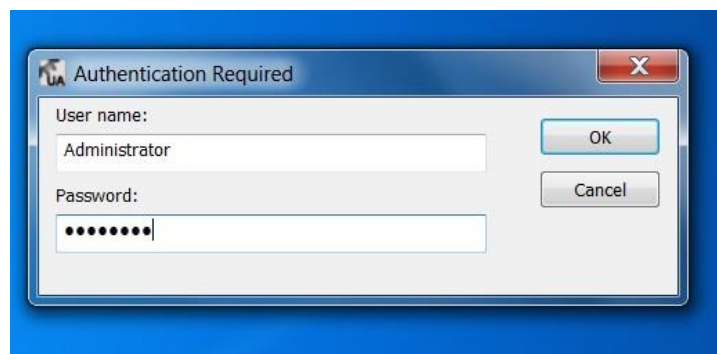
Connection Settings	
Host	127.0.0.1 Hostname or IP address of the server.
Port	49320 Port to connect to on the host.
Security Policy	Basic128Rsa15 ▾ (default: Basic128Rsa15)
Message Security Mode	SignAndEncrypt ▾ (default: SignAndEncrypt)
Username	 The username to login with.
Change Password?	☐ Check this box to change the existing password.
Password	
Password	 Re-type password for verification.

☐ Show advanced properties?

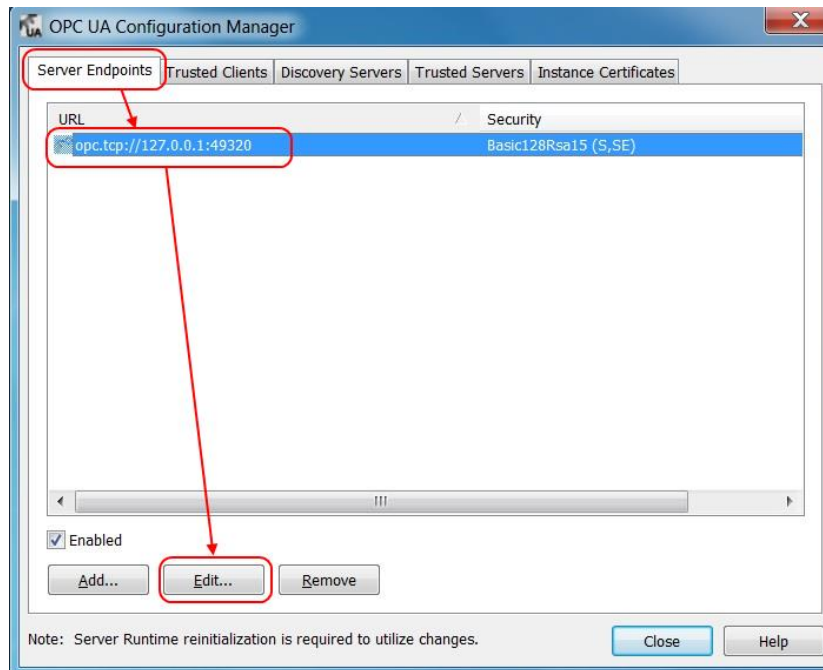
Next configure the Kepware OPC-UA server: Right click the Kepware icon in the system tray and select OPC UA Configuration.



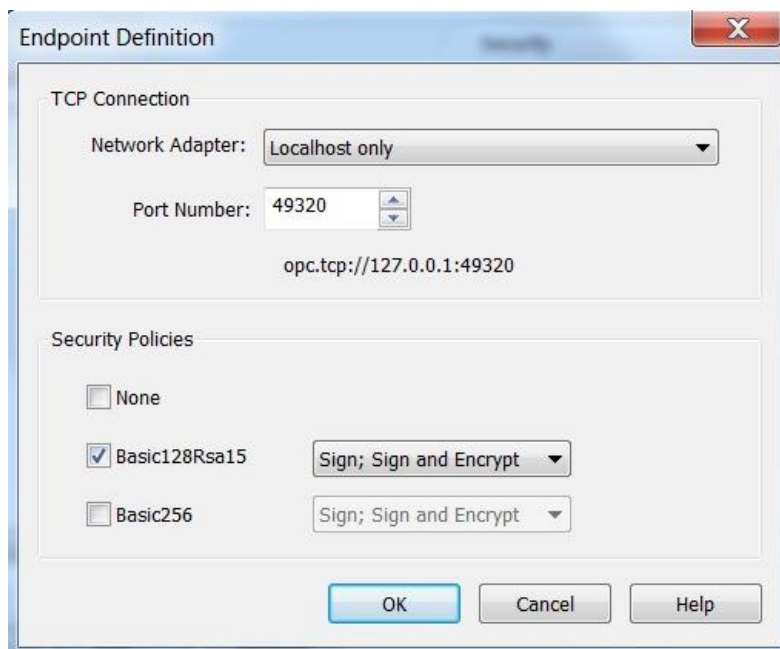
Enter user name and password – default is “Administrator” & “password”.



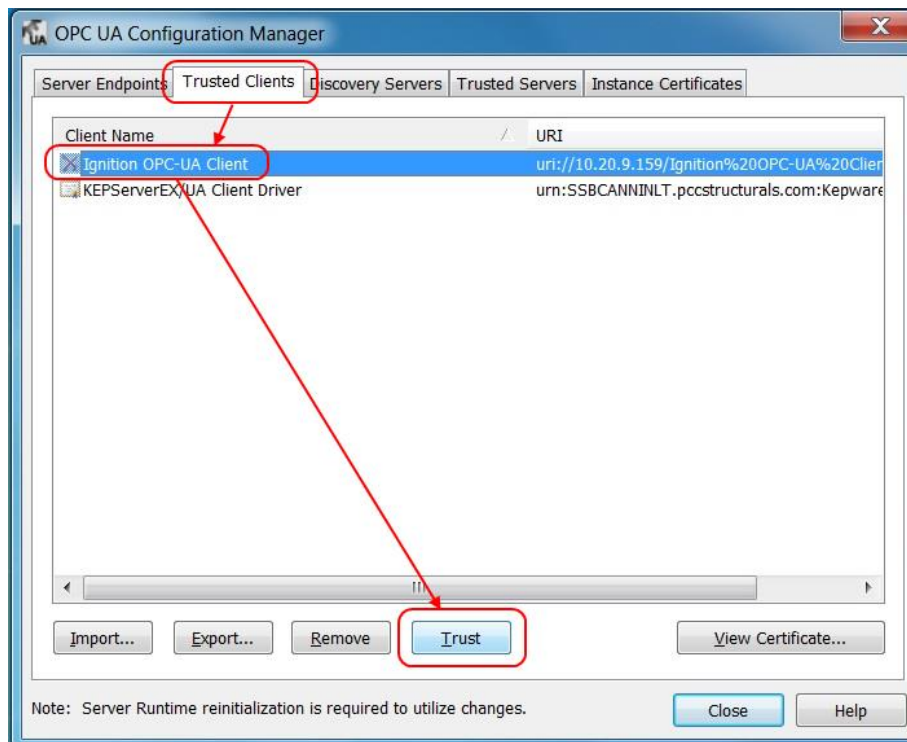
In the configuration manager select server endpoints, select the server URL and then click edit.
If there is no server URL click add.



In the edit definition screen choose the correct network adapter, leave the default port at 49320, Check “Basic 128Rsa15” and uncheck “None” and “Basic256” then click OK.



In the configuration manager select trusted clients, select the Ignition OPC-UA client then click Trust.



Now when you go back to the Ignition OPC Server Connections screen the Kepware connection should now show Connected.

System

- Status
- Licensing
- Backup/Restore
- Console
- User Manual

Configuration

- Projects
- Modules
- Gateway Settings
- Redundancy

Databases

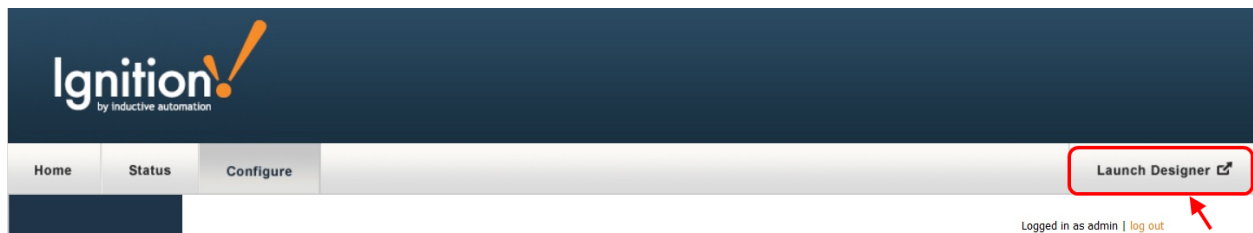
OPC Server Connections

Name	Type	Description	Status	
Ignition OPC-UA Server	OPC-UA Connection	The default connection to Ignition's OPC-UA server.	Connected	edit delete
Kepware	OPC-UA Connection	GE Fanuc ethernet demo	Connected	edit delete
Production	Production_ServerType?	Production OPC-UA connection.	Faulted	delete

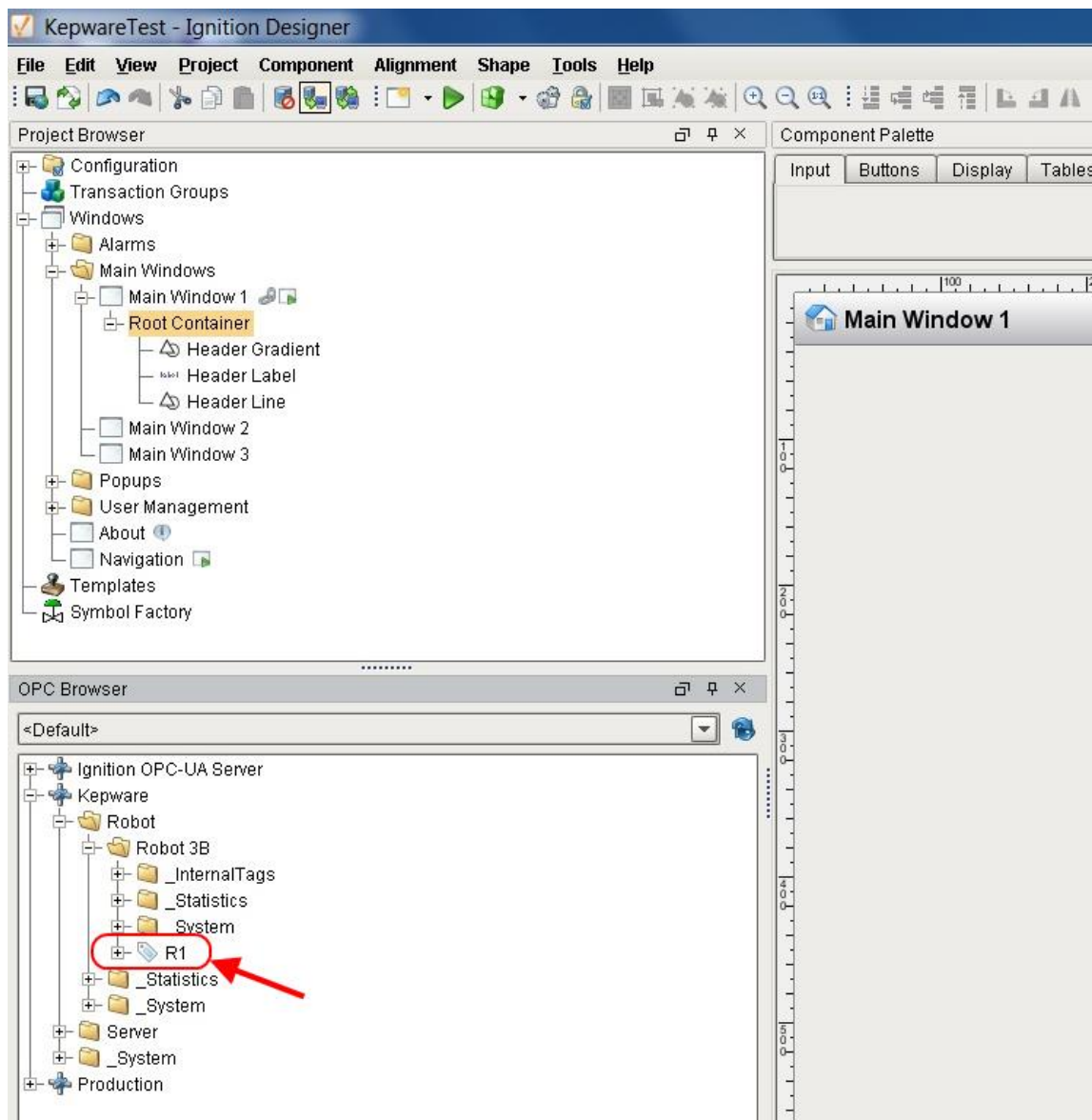
➔ [Create new OPC Server Connection...](#)

Note: For details about a connection's status, see the [OPC Connection Status](#) page.

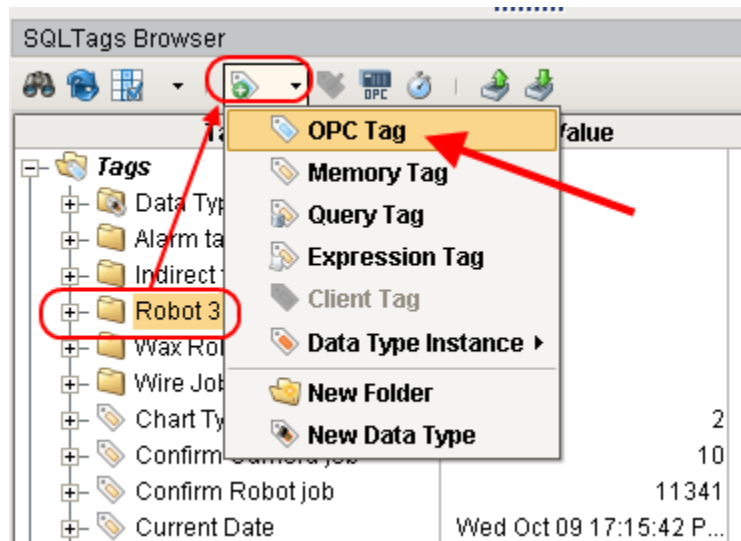
Launch the Designer.



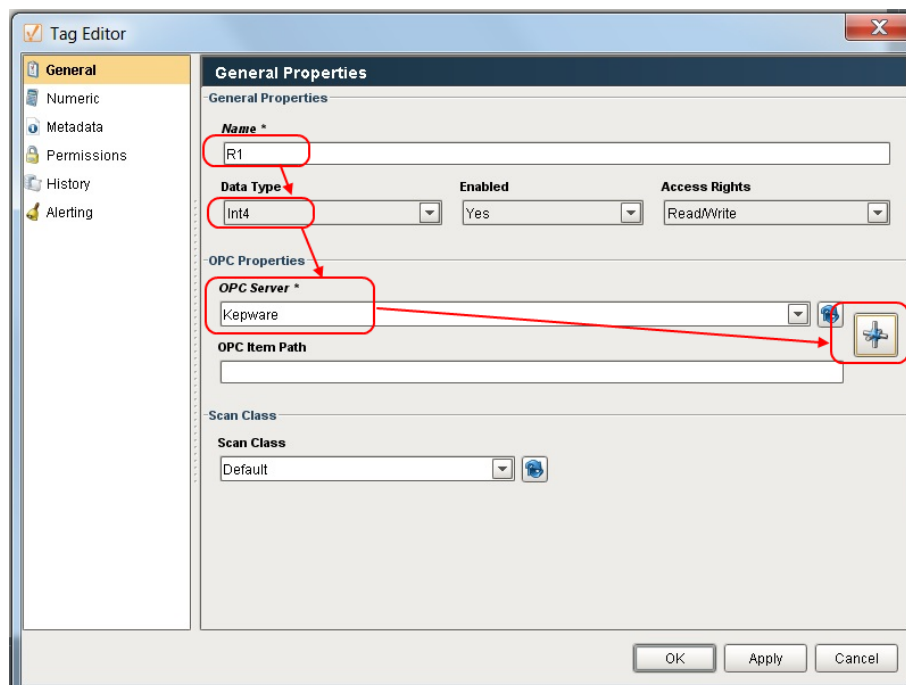
In the OPC Browser under Kepware server you should see all your new tags. (in this example I had only set up one standard register tag)



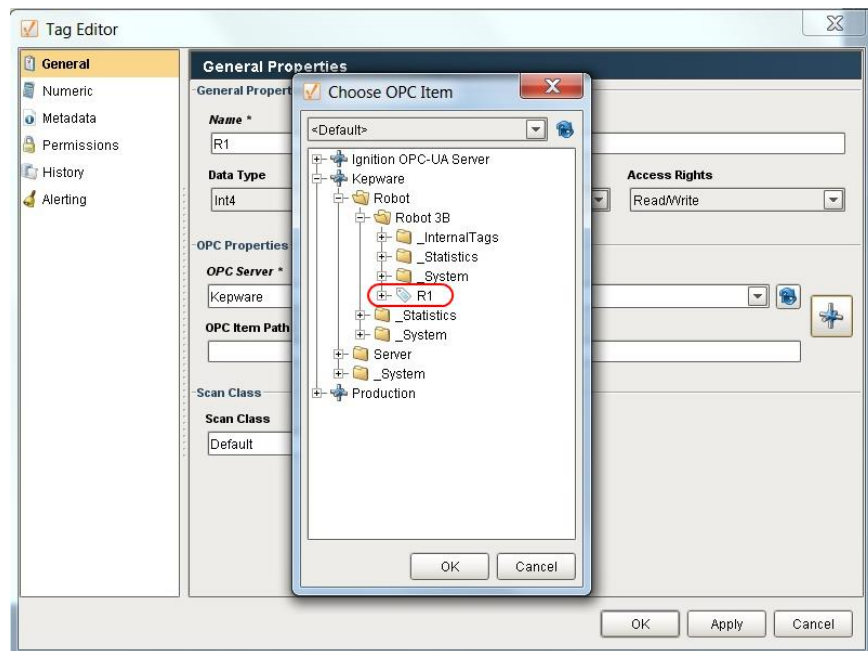
To use this tag in your project open SQL Tag Browser, create or select folder where you want the tag then from the new tag drop down choose OPC Tag.



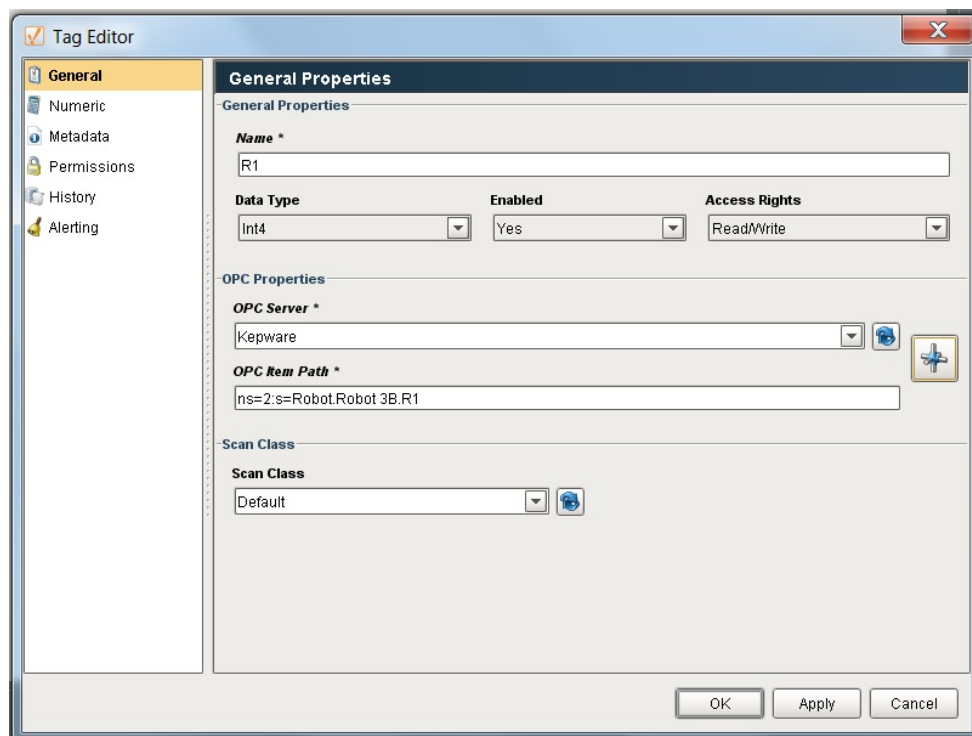
The tag editor will open, give the tag a name, select the data type then click on the OPC Item path browser button.



Navigate to and select your new tag, then click OK.



The path to your tag will now be displayed, click OK.



You can now use the tag in your program.

